

Compositional Operator Networks for Parametric, Geometry-Dependent Flow Fields

A framework for identifiable, structured latent spaces in fluid-mechanical surrogate modelling

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May 9, 2026

Abstract

Operator-learning surrogates can replace expensive PDE solves for parametric, geometry-dependent flow fields, but their latent spaces are typically opaque: high reconstruction accuracy does not imply that the learned coordinates correspond to controllable physical factors. We argue — and propose a framework to test — that fluid mechanics is a uniquely hospitable setting for identifiable, compositional latent spaces, in a way computer vision is not. Three structural facts make this possible: dimensional analysis specifies the latent axes by Buckingham’s Π -theorem; dynamic similarity guarantees that single-factor sweeps are physically realisable experiments; and the data-generating process literally factorises, because experimenters control regime and geometry independently on a designed grid. We formalise this with a block-structured latent $\mathbf{z} = [\mathbf{z}_\mu \parallel \mathbf{z}_g \parallel \mathbf{z}_\eta \parallel \mathbf{z}_\xi]$ partitioning regime, geometry, coherent dynamics, and residual fluctuations, an identifiability argument generalising the iVAE construction to operator learning, a loss library covering reconstruction, disentanglement, dynamics, physics-informed terms, and spectral-statistical statistics, and a diagnostic methodology based on linear probes, concept-vector arithmetic, and symbolic discovery on latent trajectories. We then specify three primary case studies that collectively stress every structural component — mixed convection over a heated body (geometry axis, FlowBench-based), electroconvective instabilities (multi-physics), and pitching airfoils with prescribed motion (dynamics and memory) — with a 2D Rayleigh–Bénard warm-up illustration as a clean Pi-group identifiability sanity check. The contribution is methodological: a framework, a set of falsifiable predictions, and a research programme that converts a structural claim about physics-vs-vision into testable empirical questions. The closest concurrent work demonstrates emergent semantic structure in aerodynamic latents under unstructured self-supervision; the proposal here is to make that structure architectural rather than emergent, and to test it on harder problems.

Keywords: operator learning; neural operators; disentangled representations; identifiability; symbolic discovery; reduced-order modelling; surrogate modelling; pattern formation; dynamic stall; electroconvection.

Companion document. A long-form treatment expanding every section below — including the loss library in full, the architecture survey, and the multi-objective and multidisciplinary perspectives — is maintained as a separate working document and referenced here as needed.

1 Introduction

The cost of high-fidelity fluid simulation has driven a generation of machine-learning surrogates that map parameters and geometry to flow fields: DeepONet (Lu et al., 2021), Fourier neural operators (Li et al., 2021), convolutional neural operators (Raonić et al., 2023), transformer-based

PDE solvers (Wu et al., 2024; Herde et al., 2024), implicit neural representations (Karki et al., 2025), and diffusion-style operators (Lippe et al., 2023; Nirmal et al., 2025). These methods now routinely match direct CFD on benchmark problems and are starting to enter design loops. They share an architectural assumption, however, that is rarely scrutinised: the field is everything, and the latent representation is whatever falls out of the encoder along the way.

This is a disappointing place to stop. A surrogate that reconstructs accurately but encodes the parameters of the problem into entangled, uninterpretable coordinates loses the very property that makes physical models useful for design — namely, that one can vary one knob at a time and watch the response. Modern representation-learning theory predicts this opacity outright: Locatello et al. (2019) prove that disentangled latent spaces cannot be recovered from unlabelled data without inductive bias on both the model and the data-generating process. Vision-side disentanglement work has consequently progressed by accumulating bias — β -VAE, β -TCVAE, InfoGAN, StyleGAN, iVAE, CLIP (Higgins et al., 2017; Chen et al., 2018, 2016; Karras et al., 2019; Khemakhem et al., 2020; Radford et al., 2021) — while accepting that the underlying generative process for natural images does not, in fact, factorise.

Fluid mechanics is different, and we argue this difference is structural rather than incidental. The factors of variation in a parametric flow field are not a modelling convenience: they are the dimensional groups identified by Buckingham’s Π -theorem (Buckingham, 1914), the controlled experimental knobs of dynamic similarity (Barenblatt, 1996), and the design variables of an MDO loop. Concurrent work on aerodynamic JEPAs (Giral et al., 2026) demonstrates this empirically: trained without any structural inductive bias, geometry-conditioned latents nonetheless organise so that linear probes recover flap and slat deflections at $R^2 \in [0.965, 0.988]$, and concept-vector arithmetic exposes a near-block-diagonal disentanglement matrix between flap and slat axes. That such structure emerges *by accident* on a controlled aerodynamic dataset is the strongest possible signal that it can be made *by design*.

This paper proposes the framework to do that. Our contributions are:

1. **A structural argument** (Section 2) that the obstacles preventing disentanglement in vision do not apply to fluid mechanics, with each obstacle removed by a specific physical fact: Π -theorem axes, dynamic similarity, universality classes, and factorised experimental design.
2. **A framework** (Section 3) consisting of: a block-structured latent partition; an identifiability argument generalising iVAE (Khemakhem et al., 2020) to operator learning via group-structured minibatches; a loss library spanning reconstruction, disentanglement, dynamics, physics-informed and spectral-statistical terms; and a diagnostic methodology built on linear probes, concept-vector arithmetic, and symbolic discovery on latent trajectories.
3. **Three case studies and a warm-up illustration** (Section 5) that collectively stress every structural component of the framework: a 2D Rayleigh–Bénard warm-up for clean Pi-group identifiability, mixed convection over a heated body for geometric variation (FlowBench-based), electroconvection for multi-physics generalisation, and pitching airfoils for memory and hysteresis. Each case has a concrete falsifiability target derived from existing physical theory, and each is supported by an existing dataset or a clear and feasible generation path.
4. **A research programme** (Section 7) mapping the framework onto a sequence of papers, with explicit attention to which sub-problems are tractable for undergraduate and postdoctoral collaborators.

The closest contemporary work (Giral et al., 2026) establishes empirically that aerodynamic latents organise semantically when trained with general distributional regularisation. We go further structurally (the partition is enforced rather than emergent), further dynamically (we include a propagator and rollout objectives that JEPA-style methods omit), and further physically (the case studies cover regimes — pattern formation, multi-physics, hysteresis-bearing dynamic stall — where the easier emergent-structure recipe is unlikely to generalise).

2 The compositional thesis

2.1 The setting and the problem

Let $\Omega_{\mathbf{g}} \subset \mathbb{R}^d$, $d \in \{2, 3\}$, be a spatial domain whose shape is encoded by a geometric descriptor $\mathbf{g} \in \mathcal{G}$, and let $\boldsymbol{\mu} \in \mathcal{P} \subset \mathbb{R}^p$ collect operating parameters (Reynolds number, Mach, regime intensity, forcing amplitude, ...). The space-time field of interest is

$$\mathbf{U} : \Omega_{\mathbf{g}} \times [0, T] \times \mathcal{P} \times \mathcal{G} \longrightarrow \mathbb{R}^c,$$

with c physical channels. The solution operator is

$$\mathcal{S} : \mathcal{P} \times \mathcal{G} \longrightarrow L^2(\Omega_{\mathbf{g}} \times [0, T]; \mathbb{R}^c), \quad (\boldsymbol{\mu}, \mathbf{g}) \mapsto \mathbf{U}(\cdot, \cdot; \boldsymbol{\mu}, \mathbf{g}). \quad (1)$$

A neural operator is any parametric approximation $\widehat{\mathcal{S}}_{\Theta} \approx \mathcal{S}$. The question we ask is not whether $\widehat{\mathcal{S}}_{\Theta}$ reconstructs $\mathbf{U}$ accurately — many architectures already do — but whether the latent representation through which it factors is *usable*: aligned with the physical knobs $(\boldsymbol{\mu}, \mathbf{g})$, decomposable into independent semantic blocks, and supportive of downstream design, symbolic discovery, and uncertainty quantification.

2.2 Why physics is hospitable to compositionality, where computer vision is not

The vision-side disentanglement programme (Bengio et al., 2013; Higgins et al., 2017; Chen et al., 2018; Locatello et al., 2019) confronts a hard generative obstacle: a photograph of a person is a projection from a high-dimensional process (lighting, geometry, camera optics, skin scattering) that does not itself instantiate “identity” and “pose” as independent variables. Disentanglement here is a perceptual convenience whose ground-truth axes are at best a modelling assumption. The Locatello impossibility result (Locatello et al., 2019) applies in full force: without paired or grouped supervision, no encoder-decoder pair can be expected to align its axes with such posited factors.

Fluid mechanics removes this obstacle on four independent grounds:

Pi-groups specify the latent axes. For aerodynamic flow around a body of characteristic length L at free-stream speed U in a fluid of density ρ and viscosity ν , the Buckingham II-theorem gives the regime axes by closed-form derivation: Reynolds Re , Mach M , and geometric ratios $\mathbf{g}$; for heat transfer add Prandtl Pr ; for rotation, Rossby Ro ; and so on (Barenblatt, 1996; Pope, 2000). The block partition $\mathbf{z}_{\mu} = (\text{Re}, M, \dots)$ is not posited: it is dictated by the governing equations.

Dynamic similarity makes single-factor sweeps physically realisable. Two flows with the same II-groups are scaled copies of each other — the foundational fact behind a century of wind-tunnel practice. In vision, fixing identity while varying pose has no clean ground truth; in fluid mechanics, sweeping $\mathbf{z}_{\mu}$ at fixed $\mathbf{z}_{\mathbf{g}}$ traces a line through the dimensionless-parameter space along which there are physically realisable matched flows at every point.

Universality classes bound the residual. The Kolmogorov K41 description of inertial-range turbulence (Frisch, 1995; Pope, 2000) predicts that residual fluctuations at small scales follow a small menu of universal statistics indexed by regime. The capacity of the residual block $\mathbf{z}_{\xi}$ can therefore be bounded by physical theory rather than guessed, and spectral-matching losses can target universal scaling rather than letting the network drift toward whatever the reconstruction objective makes convenient.

The data-generating process literally factorises. Experiments in fluid mechanics — and the DNS that mirrors them — are typically acquired on a designed grid: μ is set by adjusting a pump or stagnation pressure, g is set by swapping a model. Group-labelled minibatches — the auxiliary-variable construction at the heart of iVAE (Khemakhem et al., 2020) — are not a methodological imposition; they are the natural batch structure of a fluid-mechanical dataset. The identifiability obstruction of Locatello et al. (2019) is dissolved by the experimental design, not by an additional inductive bias.

The aggregate claim is that physics gives us the axes, the supervision, the residual structure, and the experimental design that vision-side disentanglement has had to invent and that natural-image data ultimately do not support. The remaining questions are not whether structured compositionality is possible, but how to architect for it cleanly and how to test it sharply.

Caveats. Three obstacles remain genuine. Discrete regime transitions (laminar-to-turbulent, attached-to-separated, subsonic-to-supersonic) violate smooth manifold structure and may require categorical regime labels alongside continuous axes. Hard orthogonality between blocks fails near transonic conditions where Re and M couple non-linearly through shock-boundary-layer interaction; soft block-Gram and HSIC penalties (Gretton et al., 2005) are the right alternative. K41 universality is approximate, with documented departures at moderate Reynolds and near walls; the residual block must absorb these rather than pretend universality is exact. None of these caveats rule out the framework; they specify where its assumptions bend.

3 The framework

3.1 Block-structured latent

We posit a latent variable

$$z = [z_\mu \parallel z_g \parallel z_\eta(t) \parallel z_\xi(t)] \in \mathbb{R}^{L_\mu} \oplus \mathbb{R}^{L_g} \oplus \mathbb{R}^{L_\eta} \oplus \mathbb{R}^{L_\xi}, \quad (2)$$

with roles: z_μ encodes regime (Π -groups); z_g encodes geometry; $z_\eta(t)$ encodes time-evolving coherent macro-scale modes; $z_\xi(t)$ encodes realisation-specific residual fluctuations. The first two blocks are time-invariant; the last two evolve under a learned propagator $\Phi_{\Delta t}$. This partition is the latent-space analogue of a generalised proper orthogonal decomposition with mean subtraction, in which the mean, modes, and residual all depend on regime — but the encoder learns the decomposition non-linearly and the disentanglement is induced by training rather than by the linearity of POD.

The encoder–propagator–decoder triple $(\mathcal{E}, \Phi, \mathcal{D})$ acts as

$$U(\cdot, 0) \xrightarrow{\mathcal{E}} z_{\text{dyn}}(0) \xrightarrow{\Phi_{\Delta t}} z_{\text{dyn}}(\Delta t) \xrightarrow{\Phi_{\Delta t}} \dots \xrightarrow{\mathcal{D}} U(\cdot, t),$$

with (μ, g) injected through static encoders $\mathcal{E}_\mu, \mathcal{E}_g$ whose outputs feed Φ as conditioning rather than evolving in time.

3.2 Identifiability via group-structured supervision

The architectural partition (2) is necessary but not sufficient: without further constraint, the four blocks can scramble information arbitrarily (Locatello et al., 2019). Our identifiability argument generalises iVAE (Khemakhem et al., 2020) to operator learning. Construct minibatches in which samples share factors: every minibatch contains groups of samples at the same μ with different g , and groups at the same g with different μ . A fusion block extracts a shared sub-code from each group:

$$z_\mu^* = \text{Fuse}\{\mathcal{E}(U_i) : \mu_i = \mu^*\}, \quad z_\xi^{(i)} = \mathcal{E}(U_i) - \Pi_\mu z_\mu^*. \quad (3)$$

The group label plays the role of the auxiliary variable u in iVAE; under standard regularity, the factorisation is identifiable up to discrete symmetries in the idealised case. The argument hinges on the experimental design supplying the auxiliary variable for free — which is exactly the property of designed fluid-mechanical datasets emphasised in Section 2.

3.3 Loss families

Training objectives are organised by role rather than enumerated exhaustively here; the companion document specifies twenty-seven specific terms with motivations. Table 1 summarises the families, what each enforces, and which latent property each supports.

Family	Role	Supports
Reconstruction (L^2 , Sobolev, BL-weighted)	Field accuracy and gradient functionals (drag, St, Nu)	P1 reconstruction
Disentanglement (Pearson, HSIC, block-Gram)	Decorrelate blocks, structurally and statistically	P2 disentanglement, P3 orthogonality
Compositional identifiability (invariance, swap, contrastive)	Make the factorisation <i>functional</i> not merely statistical	P4 equivariance
Dynamics (single-step, K -step rollout, Koopman)	Stable propagation; long-horizon statistical fidelity	P6 modal universality, exposure-bias control
Physics-informed (PDE residual, conservation, BC)	Inject governing equations as soft constraints	Physical consistency
Statistical-spectral (covariance, PSD, structure functions)	Match turbulent statistics, not just point-wise fields	P7 cascade, intermittency
Regularisation (KL, sparsity, Lipschitz)	Compactness, stability, well-posedness	P5 Lipschitz, P8 sparsity

Table 1: Loss families, each summarising several specific terms detailed in the companion document. Properties (P1–P10) refer to a list of latent-space desiderata also given there.

A few terms deserve explicit emphasis here because they are the most distinctive. The *swap-consistency* loss — $\mathbb{E}[\|\mathbf{U}(\boldsymbol{\mu}_i, \mathbf{g}_j) - \mathcal{D}(\mathbf{z}_\mu^{(i)}, \mathbf{z}_g^{(j)}, \cdot, \cdot)\|^2]$ — demands that taking the regime embedding from one sample and the geometry embedding from another produces, after decoding, the actual ground-truth flow at the cross-combination. This is a strong demand: it makes the factorisation *functional* rather than merely statistical, and it is the operator-learning analogue of Buckingham-II consistency across geometries at fixed dynamic similarity. The *spectral-matching* loss — $\int |\log E_{\hat{\mathcal{U}}}(k) - \log E_{\mathcal{U}(k)}|^2 dk$ — counteracts the documented spectral bias of plain MLPs (Rahaman et al., 2019) that systematically blur dissipative scales, by directly penalising departures from the ground-truth energy spectrum across the whole resolved range.

3.4 Architecture choices

Different operator-learning architectures are appropriate for different roles within the framework, and the framework is largely architecture-agnostic at the block level. Table 2 summarises the trade-offs; the companion document provides a longer discussion. In short: SDF-based implicit decoders (Karki et al., 2025; Sitzmann et al., 2020; Tancik et al., 2020) are the natural choice

for $\mathcal{D}$ when geometry varies; Fourier neural operators (Li et al., 2021) or convolutional neural operators (Raonić et al., 2023) are appropriate for the dynamics propagator Φ on regular grids; graph neural operators (He et al., 2024) are natural on unstructured meshes; transformers (Wu et al., 2024; Herde et al., 2024) support the joint encoder when inputs are tokenised. Diffusion-style operators (Lippe et al., 2023; Nirmal et al., 2025) are appropriate for chaotic regimes where pointwise prediction is meaningless but ensemble statistics are well-defined.

Architecture	Reg. grid	Geometry	Spectrum	Long range	Best fit role
DeepONet	no	good	weak	moderate	smooth parametric problems
FNO	yes	poor	strong	strong	periodic, dynamics propagator
CNO (U-Net)	yes	via SDF	moderate	moderate	industrial regular-grid data
GNO / MeshGNN	no	strong	weak	weak	unstructured CFD
Transformer	flexible	moderate	moderate	strong	joint encoder, foundation-scale
INR (SIREN, FF)	no	strong	with FF	weak	geometry, super-resolution, $\mathcal{D}$
Diffusion	flexible	flexible	flexible	strong	chaotic, generative

Table 2: Operator-network architectures and their natural roles within the framework. Hybrid combinations are common: e.g. GNO encoder for geometry, FNO propagator for dynamics, INR decoder for continuous query.

4 Diagnostics

A framework that promises identifiable, compositional latents must specify how to test for them. We adopt and extend three diagnostic tools that have proven informative on related representation-learning problems:

Linear probes. Following standard practice in representation learning and recently demonstrated on aerodynamic latents (Giral et al., 2026), we fit ridge regressions $y \approx W\tilde{z} + b$ from each latent block to known design or operating variables on a held-out split. High R^2 recovery indicates that the variable is linearly accessible from that block; low recovery from a block where the variable should not appear (e.g. angle of attack from z_g , which is geometry-only) is itself a positive diagnostic.

Concept-vector arithmetic. The unit-norm direction $v_k = w_k / \|w_k\|$ from the k -th probe is the latent direction associated with parameter x_k . Walking the train-mean latent along v_k and reading out all parameters via the same probe matrix produces a sensitivity matrix whose diagonal measures the intended response and whose off-diagonal measures cross-talk. Aerodynamic JEPAs (Giral et al., 2026) use this construction to demonstrate that flap and slat actuation occupy near-orthogonal subspaces in their context latent. We propose this as a standard diagnostic for the framework, applied separately to z_μ , z_g , and (where meaningful) z_η .

Symbolic discovery on latent dynamics. The strongest test of $\mathbf{z}_\eta$ is whether sparse identification of nonlinear dynamics (Brunton et al., 2016; Champion et al., 2019) applied to latent trajectories recovers known reduced-order dynamical models in tractable limits. For Rayleigh–Bénard convection (Section ??), the prediction is recovery of Newell–Whitehead–Segel-type amplitude equations near onset; for pitching airfoils (Section 5.4), recovery of Beddoes–Leishman or LESP-based dynamic-stall models. These are sharp falsifiability targets: an entangled latent will produce regression coefficients with no meaningful structure, while a successfully compositional one should produce coefficients whose dependence on $\boldsymbol{\mu}$ matches published theory.

Statistical and spectral diagnostics. Beyond probing, we evaluate energy spectra, structure functions, and integrated quantities (forces, Nusselt and Strouhal numbers, drag coefficients) computed from the decoded fields. These are gradient functionals on $\partial\Omega_{\mathbf{g}}$ and they exercise the boundary-layer-weighted and Sobolev terms of the loss library. For chaotic regimes, ensemble statistics replace pointwise comparison.

5 Three case studies and a warm-up illustration

The cases below are selected on a deliberately pragmatic principle: each is supported by data already in hand or trivially generatable by the present group and its established collaborators. Three cases form the empirical core, each stressing one dominant axis of the framework while the others provide support. A fourth, smaller case — two-dimensional Rayleigh–Bénard convection at moderate aspect ratio — serves as a warm-up illustration of clean Pi-group identifiability before the framework is taken to the geometrically and dynamically richer settings of the primary cases. Each case has at least one sharp falsifiability target derived from existing physical theory.

5.1 Warm-up: two-dimensional Rayleigh–Bénard convection

Before applying the framework to geometrically or physically rich problems it is useful to demonstrate it on the simplest possible setting in which the structural argument of Section 2 can be tested cleanly. A Boussinesq fluid layer of height H heated from below at T_h and cooled from above at T_c in two dimensions, with periodic horizontal boundaries and aspect ratio $\Gamma = L_x/H$, has only three closure groups: the Rayleigh number Ra , Prandtl number Pr , and aspect ratio Γ . The geometry block $\mathbf{z}_g = \Gamma$ is a scalar; the regime block $\mathbf{z}_\mu = [\widetilde{Ra}, \log Pr]$ is two-dimensional. Sweeping $Ra \in [Ra_c, 5Ra_c]$ across two values of Pr and a small range of Γ produces a few hundred 2D Boussinesq simulations — inexpensive to generate with off-the-shelf pseudo-spectral solvers and well-suited to undergraduate-level data generation.

The warm-up has one diagnostic role: confirm, on the simplest possible problem, that linear-probe recovery of $(\widetilde{Ra}, \log Pr, \Gamma)$ from $\mathbf{z}_\mu$ and $\mathbf{z}_g$ reaches near-perfect R^2 , and that SINDy on $\mathbf{z}_\eta(t)$ stratified by (Ra, Pr) recovers Newell–Whitehead–Segel-type amplitude equations near onset (?) with regime-dependent coefficients consistent with weakly-nonlinear theory. If the framework cannot do this on a 2D Boussinesq dataset, the more ambitious cases below are not worth attempting. We treat the warm-up as a sanity check, not a primary contribution.

A natural extension of the warm-up to spiral defect chaos at large aspect ratio (??) is left for follow-on work in collaboration with the Mukherjee group at Iowa State Mechanical Engineering, whose expertise in numerical SDC is directly relevant.

5.2 Case 1: Mixed convection over a heated body

A heated body of characteristic length L at surface temperature T_w in cold ambient T_∞ , with uniform free stream U_∞ . The body shape varies across a designed library — circular and square cylinders, ellipses, spheres, low-Reynolds aerofoil sections — encoded by signed distance functions

(Karki et al., 2025; Rabeh et al., 2025b). The closure groups are Reynolds, Richardson, and Prandtl: $\text{Re} = U_\infty L / \nu \in [10, 10^3]$, $\text{Ri} = g\beta(T_w - T_\infty)L/U_\infty^2$, and Pr . The Richardson number partitions the dynamics: $\text{Ri} \ll 1$ is forced convection, $\text{Ri} \gg 1$ free, $\text{Ri} \sim \mathcal{O}(1)$ mixed.

The compositional structure stresses geometry: $\mathbf{z}_g = E^{\text{SDF}}(\phi_g)$ is high-dimensional and produced by a coordinate-based SDF encoder (He et al., 2024; Rabeh et al., 2025a), while $\mathbf{z}_\mu = [\log \text{Re}, \text{Ri}, \log \text{Pr}]$ is modest. The most stringent test is the swap-consistency loss: take $\mathbf{z}_\mu$ from one (Re, Ri) realisation, $\mathbf{z}_g$ from a different shape, decode, and compare to the actual DNS at the cross-combination. Mixed convection is the right regime for this test — forced convection is too uncoupled, pure free convection too symmetric.

The falsifiability target is engineering: classical wake correlations $\text{St}(\text{Re})$ and $\text{Nu}(\text{Re}, \text{Pr})$ should fall out of post-hoc analysis of the trained surrogate without being trained on directly, since they are gradient functionals on $\partial\Omega_g$ that depend on the Sobolev and boundary-layer-weighted losses.

5.3 Case 2: Electroconvective instabilities

A binary symmetric electrolyte fills a planar channel between two ion-selective membranes held at potentials $\pm V/2$. The governing system couples Poisson–Nernst–Planck charge transport with the incompressible Navier–Stokes equations through electrostatic body forces (?). Above a critical applied voltage the extended space-charge layer becomes hydrodynamically unstable, driving electroconvective vortices into the over-limiting current regime (??); at higher voltage the flow becomes spatio-temporally chaotic (?).

The closure groups are the dimensionless voltage $\tilde{V} = V/(RT/F)$, the dimensionless Debye length $\tilde{\lambda}_D = \lambda_D/H$, the channel aspect ratio, and Schmidt number. The compositional structure assigns $\mathbf{z}_\mu = [\tilde{V}, \log \tilde{\lambda}_D]$, $\mathbf{z}_g = \text{AR}$, with $\mathbf{z}_\eta$ encoding coherent EC vortex modes and $\mathbf{z}_\xi$ the chaotic regime above onset. The state vector now carries six channels rather than four: $(\mathbf{u}, p, c_+, c_-, \Phi)$, and the diffuse charge layer enforces extreme scale separation between the Debye scale and the channel scale.

The falsifiability target is an onset surface: the critical voltage $\tilde{V}_c(\tilde{\lambda}_D, \text{AR})$ should match Rubinstein–Zaltzman’s linear-stability prediction in the small- $\tilde{\lambda}_D$ asymptotic limit (?). This is the single piece of evidence most informative about whether the framework’s structural inductive bias generalises beyond Navier–Stokes proper.

5.4 Case 3: Pitching airfoils and dynamic stall

A NACA airfoil undergoing prescribed pitching motion $\alpha(t) = \alpha_0 + \alpha_{\text{amp}} \sin(\omega t)$ (or constant-rate pitch-up $\dot{\alpha} = \text{const}$) at chord-based Reynolds number $\text{Re} = U_\infty c / \nu$, with reduced frequency $k = \omega c / (2U_\infty)$ and pitch axis at x_p/c . The flow exhibits a leading-edge vortex (LEV) that forms, grows, sheds, and modulates the unsteady lift; for sufficiently aggressive pitch rates the system enters dynamic stall, characterised by lift overshoot during pitch-up, deep separation during pitch-down, and a closed hysteresis loop in (C_L, C_M) space (Sharma and Visbal, 2019; Sudharsan et al., 2022; Sudharsan and Sharma, 2025). Existing high-resolution LES databases at multiple thicknesses, Reynolds numbers, and pitch rates — collected and curated by collaborators at Iowa State (Sharma and Visbal, 2019; Sudharsan et al., 2022) — provide the training data, with an established collaboration on dynamic-stall criteria already in place between the present group and the Sharma group.

The compositional structure here tests *dynamics and memory*, the only one of the four cases that does. The latent partition is $\mathbf{z}_\mu = [\log \text{Re}, \log k, \alpha_0, \alpha_{\text{amp}}, x_p/c]$, $\mathbf{z}_g = \text{airfoil-shape SDF embedding}$, $\mathbf{z}_\eta(t) = \text{LEV trajectory and phase}$, $\mathbf{z}_\xi(t) = \text{transitional fluctuations during stall}$. Because the response at angle α during pitch-up differs systematically from the response at α during pitch-down — the system is hysteretic — the propagator Φ must track *history*, not just state. This is exactly the regime where naive autoregressive rollouts fail in interesting ways and

where the rollout, static-invariance, and Koopman-spectral terms of the loss library earn their keep.

The falsifiability target is reduced-order: SINDy on $\mathbf{z}_\eta$ stratified by (k, α_0) should recover Beddoes–Leishman-type or LESP-based dynamic-stall models with regime-dependent coefficients, and should reproduce the vorticity-based onset criterion of [Sudharsan et al. \(2022\)](#). The integrated lift coefficient $C_L(\alpha; k, \text{Re})$ recovered from the decoded fields should reproduce published hysteresis loops to within experimental scatter.

5.5 What the three cases collectively certify

	Case 1: Heated body	Case 2: Electroconvection	Case 3: Pitching airfoil
Stresses	geometry $\mathbf{z}_g$	multi-physics	dynamics, memory
Π -groups in $\mathbf{z}_\mu$	$\log \text{Re}, \text{Ri}, \log \text{Pr}$	$\tilde{V}, \log \tilde{\lambda}_D$	$\log \text{Re}, \log k, \alpha_0, \alpha_{\text{amp}}$
$\mathbf{z}_g$ dimension	high-D SDF	scalar (AR)	moderate-D SDF
Channels	4	6–7	4
Symbolic target	St, Nu scalings	onset $\tilde{V}_c$	dynamic-stall ROM, vorticity criterion
Memory effect	none (steady wake)	weak (chaotic)	strong (hysteresis)
Data status	FlowBench in hand, extensible (Tali et al., 2025)	moderate generation effort (?)	Sharma archives in hand (Sharma and Visbal, 2019)

Table 3: The three primary case studies, organised by which framework component each stresses. The warm-up illustration of Section 5.1 provides a clean Pi-group identifiability sanity check on a simpler 2D problem.

The cases progress from a clean low-dimensional regime sweep on a simple 2D problem (warm-up) through complex geometry at moderate regime variation (Case 1), multi-physics at fixed simple geometry (Case 2), and genuinely unsteady prescribed-motion problems with memory (Case 3). Each case has independent existing literature against which the framework’s predictions can be checked; none requires speculative new physics. This makes the empirical programme cleanly falsifiable: a failure on the warm-up localises to basic Pi-group identifiability, a failure on Case 1 to geometry encoding, on Case 2 to multi-physics generalisation, and on Case 3 to memory and hysteresis tracking. The progression is also data-pragmatic: each primary case is supported by an existing dataset or a clear and feasible generation path, with no critical-path dependence on collaborations not yet in place.

6 Relation to concurrent work

The closest contemporary work to this proposal is AeroJEPA ([Giral et al., 2026](#)), which introduces a JEPA-style ([Assran et al., 2023](#); [Balestrieri and LeCun, 2025](#)) predictive latent architecture for 3D aerodynamic surrogate modelling, with SIGReg as its collapse-avoidance regulariser and an INR decoder for continuous query. Trained without any structural inductive bias on flap/slat or operating-condition labels, AeroJEPA’s context latents are nonetheless found, after training, to support linear probing of high-lift control-surface deflections at $R^2 \in [0.965, 0.988]$, near-block-diagonal disentanglement between flap and slat axes via concept-vector arithmetic, and a proof-of-concept gradient-based optimisation in latent space.

We view AeroJEPA as concurrent empirical evidence for the structural thesis of Section 2: when the data-generating process literally factorises (controlled flap and slat sweeps over a

designed grid of geometries and operating conditions), even unstructured self-supervised training discovers approximately disentangled representations. This is consistent with the central claim here, and with the iVAE identifiability argument it generalises.

The differences are four. *First*, the disentanglement reported in (Giral et al., 2026) is emergent, while ours is enforced — we provide a structural partition with explicit losses targeting block independence and an iVAE-style identifiability argument, where AeroJEPA relies on SIGReg plus reconstruction. *Second*, AeroJEPA is steady-state surrogate plus linear probes; it has no propagator, no rollout objective, and no exposure-bias machinery. The cases where compositionality earns its scientific keep — chaotic flows, regime transitions, hysteretic dynamic stall, symbolic discovery from latent trajectories — are outside its current scope. *Third*, AeroJEPA contains no Π -group identification, no physics-informed losses, no spectral-statistical losses, and no scale separation; its flows are clean, attached, smooth surface fields on transonic wings. *Fourth*, the diagnostic tools introduced in (Giral et al., 2026) — linear probing and concept-vector arithmetic — transfer cleanly into our framework as evaluation methodology, and we adopt them here.

In short, AeroJEPA validates the easier version of the claim: that aerodynamic latents organise semantically when trained well. The harder claim — that one can architect for compositionality and have it survive into chaotic, multi-physics, and hysteretic regimes — is what this programme tests.

7 Research programme

The framework, the three primary cases, the warm-up illustration, and the diagnostic methodology together specify a sequence of empirical questions. Each natural deliverable corresponds to a publishable unit, and the order below reflects data availability rather than conceptual ordering: cases supported by existing or trivially generatable data come first, with the more ambitious extensions sequenced behind them.

Paper 1 (anchor): Framework + warm-up + Case 1. The present document, with full empirical demonstration on (i) the 2D Boussinesq warm-up of Section 5.1 as a clean Π -group identifiability sanity check, and (ii) Case 1 (mixed convection over heated bodies, FlowBench-based) as the primary empirical case. Deliverables: framework, identifiability argument, full loss library, diagnostic methodology, ablation studies of the loss components, and complete results on the warm-up and Case 1 including linear-probe recovery of design and operating variables, concept-vector arithmetic, and Strouhal/Nusselt recovery from decoded fields. Target venue: a journal that takes both ML and fluid-mechanics readers (*J. Comput. Phys.*, *Nature Communications*, or a focused ML/physics venue). Lead: postdoc.

Paper 2: Case 3 (dynamics and memory). Pitching airfoils with prescribed kinematic schedules, stressing the propagator, rollout objectives, and hysteresis. Builds on the existing Sudharsan et al. (2022) collaboration on dynamic-stall criteria and leverages Sharma’s pitching-airfoil LES databases (Sharma and Visbal, 2019). The natural falsifiability target is recovery, by SINDy on z_η , of the vorticity-based dynamic-stall onset criterion of Sudharsan et al. (2022) — a self-validation of the framework against the group’s own published reduced-order theory. Target venue: *J. Fluid Mech.* or *AIAA Journal*. Lead: undergraduate + postdoc, in collaboration with Sharma’s group.

Paper 3: Case 2 (multi-physics). Electroconvective instabilities, stressing multi-channel reconstruction, scale separation, and onset prediction. Builds on the Iowa State EC-DNS infrastructure (?). The challenge here is the moderate generation effort for the training-data sweep, partly mitigated by the framework’s tolerance for physics-informed regularisation as a

substitute for data abundance. Target venue: *Phys. Rev. Fluids*, *J. Fluid Mech.*, or *J. Comput. Phys.* Lead: postdoc with chemistry collaborators.

Paper 4 (stretch): Pattern formation and spiral defect chaos. A full extension of the warm-up to large-aspect-ratio Rayleigh–Bénard convection, including the spiral defect chaos regime and the secondary-instability cascade (??). Naturally collaborative with the Mukherjee group at Iowa State Mechanical Engineering, whose expertise in numerical SDC (e.g. [Mukherjee and Paul, 2022](#)) provides both the simulation infrastructure and the theoretical context. The falsifiability target is amplitude-equation recovery via SINDy on $\mathbf{z}_\eta$ across the bifurcation cascade. Target venue: *J. Fluid Mech.* or a pattern-formation specialist venue. Lead: postdoc + Mukherjee.

Methodological side papers. As the empirical programme matures, two methodological papers naturally arise. The first is a formalisation of the iVAE-to-operator-learning identifiability argument (theory paper, lead by postdoc with mathematics collaborators). The second is a focused study of symbolic discovery on latent trajectories with sparsity-constrained regression ([Champion et al., 2019](#)), demonstrating the recovery of dynamic-stall reduced-order models (Case 3) and amplitude equations (Paper 4) on real latent trajectories.

Tractability for the team. The 15-week first-semester project plan accompanying this document scopes the anchor paper’s empirical content into a structured loss-ablation study suitable for a postdoc plus two undergraduate collaborators, with light involvement from the Sharma group ramping toward Paper 2. Two sub-projects in particular are well-suited for undergraduate ownership under postdoc supervision: the diagnostic suite (linear probes, concept-vector arithmetic, ridge-regression-based design-space exploration) applied to baseline and compositional encoders on the warm-up; and the geometry-library construction and physics-informed loss implementation for Case 1, which is a self-contained data-pipeline project that feeds directly into the anchor paper.

8 Open questions

Several questions are flagged by the framework but remain open. In which regimes is the propagator Φ approximately linear on a Koopman-invariant subspace of $\mathcal{Z}$, and where must it be genuinely non-linear? Can the iVAE-style identifiability argument of Section 3 be tightened into a theorem for the operator-learning setting under explicit regularity conditions on the data-generating process? For chaotic flows, the residual block $\mathbf{z}_\xi$ is genuinely correlated with shedding events in $\mathbf{z}_\eta$, so hard orthogonality between blocks is too strong — what is the right soft-orthogonality penalty, and how should its weight be set adaptively as the flow approaches transition? How does conditional-entropy sizing of L_ξ track $H(\mathbf{U} \mid \boldsymbol{\mu}, \mathbf{g})$ across the cases, and is the dimensionality predictable from physical theory? The answer to each of these questions is a paper in itself.

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soft-orthogonality penalty, and how should its weight be set adaptively as the flow approaches transition? How does conditional-entropy sizing of L_ξ track $H(\mathbf{U} \mid \boldsymbol{\mu}, \mathbf{g})$ across the four cases, and is the dimensionality predictable from physical theory? The answer to each of these questions is a paper in itself.

Companion documents. The full mathematical framework, the loss library in detail, the architecture survey, the multi-objective and multidisciplinary perspectives, and extended commentary boxes for both fluid-mechanics and ML readerships are maintained in a separate working document available on request. The case-specific draft sections are similarly developed in companion files.

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